

4

Spatial point processes

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4.1 Processes and summary descriptions

A *spatial point process* is a stochastic mechanism that generates a countable set of events x_i in the plane. We will usually be dealing with processes that are *stationary* and *isotropic*. Stationarity means that all properties of the process are invariant under translation, isotropy that they are invariant under rotation. These assumptions are less restrictive than they might seem at first sight. In particular, they do not rule out the modelling of *random* heterogeneity in the environment (recall the discussion of Figure 1.2 in Section 1.1). However, three qualifications are in order.

Firstly, although models are often defined as processes on the whole plane, in practice we only apply them to data from finite planar regions and it will be sufficient for our purposes if stationarity and isotropy hold to a reasonable approximation within the study region in question. Indeed, study regions are

often selected with this requirement implicitly in mind so that, for example, in micro-anatomical studies the tissue sections to be analysed are deliberately sited well away from any boundaries between different types of tissue.

Secondly, we will abandon the stationarity assumption when the data include spatial explanatory variables that account for spatial variation in the local intensity of events.

Finally, in some applications we may choose to circumvent the stationarity assumption by the use of design-based inference. This applies to the analysis of replicated patterns as discussed in Chapter 5, to certain problems in environmental epidemiology which we consider in Chapter 9, and to our discussion of mechanistic models for spatio-temporal point processes in Chapter 13.

Many statistical methods for spatial point pattern data involve comparisons between empirical summary descriptions of the data and the corresponding theoretical summary descriptions of a point process model. However, it is important that the theoretical summary descriptions are derived from an underlying model, rather than being advanced as models in their own right. For example, we have already seen in Chapter 2 that one summary description of the homogeneous Poisson process is that the distribution function of the distance from an arbitrary event of the process to its nearest neighbour is

$$G(y) = 1 - \exp(-\lambda\pi y^2) : y \geq 0.$$

This leads, amongst other things, to the construction of tests of complete spatial randomness involving a comparison between this theoretical form of $G(y)$ and the corresponding empirical distribution function for an observed pattern of n events. However, it would make no sense to attempt to define a more general class of models by embedding $G(y)$ within a larger parametric family of distributions unless the enlarged class were itself derived from an explicit class of point process models.

In this chapter, we consider various theoretical summary descriptions of point processes, and the corresponding empirical descriptions of point pattern data. We focus on properties that lead to useful statistical methods, and illustrate their use on a number of data-sets. We include a description of the homogeneous Poisson process and, for multivariate processes, discuss the ideas of independence and random labelling. We postpone until Chapter 6 a general discussion of different parametric classes of point processes that have been proposed as models for data.

We need the following notation: $E[X]$ denotes the expectation of a random variable X ; $N(A)$ denotes the number of events in the planar region A ; in a multivariate process, $N_j(A)$ similarly denotes the number of type j events in A ; $|A|$ is the area of A ; dx is an infinitesimal region that contains the point x ; $\|x - y\|$ denotes the Euclidean distance between the points x and y .

4.2 Second-order properties

4.2.1 Univariate processes

We can now define the *first-order* and *second-order* properties of a spatial point process. First-order properties are described by an *intensity function*,

$$\lambda(x) = \lim_{|dx| \rightarrow 0} \left\{ \frac{E[N(dx)]}{|dx|} \right\}.$$

For a stationary process, $\lambda(x)$ assumes a constant value λ , the mean number of events per unit area.

The *second-order intensity function* is similarly defined as

$$\lambda_2(x, y) = \lim_{|dx|, |dy| \rightarrow 0} \left\{ \frac{E[N(dx)N(dy)]}{|dx||dy|} \right\}.$$

A closely related quantity is the *conditional intensity* $\lambda_c(x|y) = \lambda_2(x, y)/\lambda(y)$ which, loosely speaking, corresponds to the intensity at the point x conditional on the information that there is an event at y .

For a stationary process, $\lambda_2(x, y) \equiv \lambda_2(x - y)$; for a stationary, isotropic process, $\lambda_2(x - y)$ reduces further to $\lambda_2(t)$, where $t = \|x - y\|$. In statistical mechanics, the scaled function $\rho(t) = \lambda_2(t)/\lambda^2$ is referred to as the *radial distribution function*, or *pair correlation function*, although it is neither a distribution function nor a correlation function in the usual statistical sense.

Baddeley, Møller and Waagepetersen (2000) discuss *intensity-reweighted (second-order) stationary* processes, which have the property that

$$\lambda_2(x, y)/\lambda(x)\lambda(y) = \rho(t) \tag{4.1}$$

depends only on $t = \|x - y\|$. Note that this requires the intensity function $\lambda(x)$ to be bounded away from zero, in which case we again call $\rho(t)$ the pair correlation function. Intensity-reweighted stationarity is a point process analogue of the assumption commonly made in the analysis of real-valued spatial processes that the mean value may vary spatially whereas the variation about the local mean is stationary.

An alternative characterisation of the second-order properties of a stationary, isotropic process is provided by the function $K(t)$, one definition of which is

$$K(t) = \lambda^{-1}E[N_0(t)], \tag{4.2}$$

where $N_0(t)$ is the number of further events within distance t of an arbitrary event. The notion of an arbitrary event of the process involves the conceptual limit of simple random sampling from a finite population. For a mathematically rigorous discussion see, for example, Daley and Vere-Jones (2002, 2005). Intuitively, we envisage a large but finite number n of events in some finite

region and define an arbitrary event to be an event selected at random from this population. Similarly, in practice an arbitrary point will mean a point distributed uniformly over some finite region.

In order to establish a link between $K(t)$ and $\lambda_2(t)$ we shall assume that our process is *orderly*, by which we mean that multiple coincident events cannot occur or, more precisely, that $P\{N(dx) > 1\}$ is of a smaller order of magnitude than $|dx|$. This means that $E[N(dx)] \sim P\{N(dx) = 1\}$ in the sense that the ratio of these two quantities tends to 1 as $|dx| \rightarrow 0$. We shall further assume that in a similar sense, $E[N(dx)N(dy)] \sim P\{N(dx) = N(dy) = 1\}$. Under these conditions, the expected number of further events within distance t of an arbitrary event can be computed by integrating the conditional intensity over the disc with centre the origin and radius t . Hence,

$$\lambda K(t) = \int_0^{2\pi} \int_0^t \{\lambda_c(x|o) x dx d\theta.$$

Using the fact that $\lambda_c(x|o) = \lambda_2(x)/\lambda$, this gives

$$\lambda K(t) = 2\pi\lambda^{-1} \int_0^t \lambda_2(x) x dx, \quad (4.3)$$

or conversely, $\lambda_2(t) = \lambda^2(2\pi t)^{-1} K'(t)$ and

$$\rho(t) = (2\pi t)^{-1} K'(t). \quad (4.4)$$

Note that for an intensity-reweighted stationary process, Baddeley, Møller and Waagepetersen (2000) extend the definition of the K -function to

$$K_I(t) = 2\pi \int_0^t \rho(x) x dx,$$

which reduces to (4.3) in the stationary case.

It is sometimes more convenient to work with $\lambda_2(t)$ or $\rho(t)$ rather than with $K(t)$. As a minor variation we define a *covariance density*,

$$\gamma(t) = \lambda_2(t) - \lambda^2 = \lambda^2 \{\rho(t) - 1\}. \quad (4.5)$$

For data analysis, one potential advantage of $K(t)$ over $\lambda_2(t)$ or $\rho(t)$, especially in small samples, is that its estimation is more straightforward. Essentially, $K(t)$ and $\lambda_2(t)$ are related to the distribution function and probability density function of the distances between pairs of events in a point pattern, and the former can be estimated without having to decide how much to smooth the corresponding empirical distribution. Against this, given a data-set containing sufficiently many events, a simple histogram-like estimate of $\lambda_2(t)$ or $\rho(t)$ is easily calculated and may be considered easier to interpret. We return to this point in Section 4.6.2.

Another useful property of the K -function is that it is invariant under

random thinning. By this, we mean that if each event of a process is retained or not according to a series of mutually independent Bernoulli trials, **then the K -function of the resulting thinned process is identical to that of the original, unthinned process.** This follows from the definition (4.2), where the K -function is defined as the ratio of two quantities, $E[N_0(t)]$ and λ . The effect of the thinning is to multiply each of these by p , the retention probability for any one event, leaving the ratio unchanged.

Rather than observe the exact locations of events in a planar region, it is sometimes easier in practice to observe only counts $N(A)$ in convenient sub-regions A (cf. Section 2.7). The resulting *quadrat count distribution*,

$$p_n(A) = P\{N(A) = n\} : n = 0, 1, \dots,$$

provides a possible summary description of the process. The arbitrary nature of A is unsatisfactory. One solution is further to summarize the quadrat count distribution by its first few moments, and to regard these as functions of A . In particular,

$$E[N(A)] = \int_A \lambda(x) dx,$$

which reduces to $\lambda|A|$ for a stationary process. More interestingly, orderliness implies that

$$\begin{aligned} E[N(A)^2] &= E\left[\left\{\int_A N(dx)\right\}^2\right] \\ &= E\left[\int_A N(dx) + \int_A \int_A N(dx)N(dy)\right]. \end{aligned}$$

Interchanging expectation and integration this becomes, for a stationary process,

$$\lambda|A| + \int_A \int_A \lambda_2(x-y) dx dy,$$

whence

$$\text{Var}\{N(A)\} = \int_A \int_A \lambda_2(x-y) dx dy + \lambda|A|(1 - \lambda|A|). \quad (4.6)$$

A straightforward generalization gives

$$\text{Cov}\{N(A), N(B)\} = \int_A \int_B \lambda_2(x-y) dx dy + \lambda|A \cap B| - \lambda^2|A||B|$$

where $A \cap B$ denotes the intersection of A and B .

The quantity $\text{Var}\{N(A)\}$ defined in (4.6) is sometimes called the “variance-area curve”, and is closely related to $K(t)$ in the sense that both involve integrated versions of the second-order intensity function.

4.2.2 Extension to multivariate processes

In a *multivariate* process, the events are of two or more distinguishable types. Definitions for the second-order properties of such processes follow as natural generalisations of the corresponding quantities for univariate processes. We assume stationarity, isotropy and orderliness, and write $N_j(S)$ for the number of type j events in a planar region A .

The (*first-order*) *intensities* are constants,

$$\lambda_j = E[N_j(A)]/|A|.$$

The *second-order intensities* are functions of a scalar argument,

$$\lambda_{ij}(t) = \lim_{\substack{|dx| \rightarrow 0 \\ |dy| \rightarrow 0}} \left\{ \frac{E[N_i(dx)N_j(dy)]}{|dx||dy|} \right\}$$

where, as before, t denotes distance. The corresponding *covariance densities* are

$$\gamma_{ij}(t) = \lambda_{ij}(t) - \lambda_i \lambda_j$$

and the multivariate K -functions are

$$K_{ij}(t) = \lambda_j^{-1} E[N_{0ij}(t)] \quad (4.7)$$

where $N_{0ij}(t)$ denotes the expected number of (further) type j events within distance t of an arbitrary type i event. Note that $\lambda_{ij}(t) = \lambda_{ji}(t)$, from which it follows that $\gamma_{ij}(t) = \gamma_{ji}(t)$. A similar argument to the one used in Section 4.2.1 shows that

$$K_{ij}(t) = 2\pi(\lambda_i \lambda_j)^{-1} \int \lambda_{ij}(x) x dx$$

from which it follows that $K_{ji}(t) = K_{ij}(t)$.

4.3 Higher order moments and nearest neighbour distributions

Second-order properties provide a natural and valuable starting point for the description of a spatial point process. However, they do not give a complete picture. Baddeley and Silverman (1984) describe a class of non-Poisson processes for which $K(t) = \pi t^2$, and in Section 6.9.3 we shall give another example in which clearly different processes have identical second-order properties. Higher order properties can easily be defined in terms of the joint intensity functions for the occurrence of specified configurations of three, four, etc. events. Interpretation would be difficult in practice since, for example, the

third-order intensity function of a stationary, isotropic process requires three arguments, the fourth-order function five, and so on.

In view of this, we define two distribution functions that we used in Chapter 2 to provide tests of CSR and that serve as additional summary descriptions for spatial point processes. These are $G(y)$, the distribution function of the distance from an arbitrary *event* to the nearest other event, and $F(x)$, the distribution function of the distance from an arbitrary *point* to the nearest event.

One interpretation of $F(x)$ is as the probability that a randomly located disc of radius x contains at least one event. This suggests an obvious extension whereby the disc is replaced by some other geometrical figure. Oriented shapes such as ellipses or rectangles could be used to describe departures from isotropy. Matheron (1975) incorporates these ideas within a general theory of random sets, in which a random set S is characterized by the function $\mathcal{F}(\mathcal{T})$, the probability that the intersection of S with T is non-empty, for a suitably wide class of “test-sets” T .

4.4 The homogeneous Poisson process

The homogeneous planar Poisson process, subsequently referred to without qualification as the Poisson process, is the cornerstone on which the theory of spatial point processes is built. It represents the simplest possible stochastic mechanism for the generation of spatial point patterns, and in applications is used as an idealized standard of complete spatial randomness that, if strictly unattainable in practice, sometimes provides a useful approximate description of an observed pattern. The Poisson process is conveniently defined by the following postulates, which correspond exactly to the definition of complete spatial randomness given in Section 1.3:

PP1 For some $\lambda > 0$, and any finite planar region A , $N(A)$ follows a Poisson distribution with mean $\lambda|A|$.

PP2 Given $N(A) = n$, the n events in A form an independent random sample from the uniform distribution on A .

To demonstrate that PP1 and PP2 are self-consistent, we first establish

PP3 For any two disjoint regions A and B , the random variables $N(A)$ and $N(B)$ are independent.

Let $C = A \cup B$ be the union of two disjoint regions A and B . Write $p = |A|/|C|$ and $q = 1 - p = |B|/|C|$. Then, PP2 applied to the region C implies that

$$P\{N(A) = x, N(B) = y | N(C) = n\} = \binom{x+y}{x} p^x q^y,$$

for integers $0 \leq x \leq n$ and $y = n - x$. PP1 then gives the unconditional joint distribution of $N(A)$ and $N(B)$ as

$$\begin{aligned} P\{N(A) = x, N(B) = y\} &= \binom{x+y}{x} p^x q^y \{e^{-\lambda|C|} (\lambda|C|)^n / n!\} \\ &= \{e^{-\lambda|A|} (\lambda|A|)^x / x!\} \{e^{-\lambda|B|} (\lambda|B|)^y / y!\} \end{aligned} \quad (4.8)$$

for all integers $x \geq 0$ and $y \geq 0$. This establishes PP3 and shows also that $N(A)$ and $N(B)$ have the distributions implied by PP1. It is immediately obvious that if PP2 holds for any region C it must hold also for all sub-regions of C . Conversely, the additive property of independent Poisson-distributed random variables X and Y , and the associated conditional binomial distribution of X given $X + Y$, establish PP1 and PP2 respectively for any region formed as the union of two disjoint regions for which PP1 and PP2 hold. This proves the required self-consistency.

The parameter λ of the Poisson process is its intensity. The independence result PP3 implies that

$$\lambda_2(t) = \lambda^2 : t > 0, \quad (4.9)$$

whence (4.9) gives

$$K(t) = \pi t^2 : t > 0. \quad (4.10)$$

The variance-area curve follows directly from PP1 as

$$\text{Var}\{N(A)\} = \lambda|A|. \quad (4.11)$$

The nearest neighbour distribution functions $G(y)$ and $F(x)$ are identical, since the existence of an event at a particular point, x_0 say, has no bearing on the distribution of the remaining number of events in a disc with centre x_0 . We deduce from PP1 that

$$F(x) = G(x) = P\{N(\pi x^2) > 0\} = 1 - \exp(-\pi \lambda x^2) : x > 0. \quad (4.12)$$

To simulate a partial realisation of a Poisson process on A conditional on a fixed value of $N(A)$, we need to generate events independently according to a uniform distribution on A . Awkward shapes of region can be accommodated by simulating the process on a larger region of a more convenient shape, such as a rectangle or disc, and retaining only those events which lie within A . Alternatively, Hsuan (1979) gives an algorithm for the direct generation of events uniformly distributed on an arbitrary polygon.

If $N(A)$ is required to be randomly varying, this same method can of course be preceded by the simulation of $N(A)$ from the appropriate Poisson distribution. In some implementations, the direct simulation of $N(A)$ is a relatively time-consuming step. Lewis and Shedler (1979) propose an alternative method which can be used when A is a rectangle, say $A = (0, a) \times (0, b)$. This is based on the observation that the x -coordinates of events in the infinite strip

$0 \leq y \leq b$ form a *one-dimensional* Poisson process with intensity λb , from which it follows that the differences between successive ordered x -coordinates are independent realizations of an exponentially distributed random variable with distribution function

$$F(v) = 1 - e^{-\lambda bv}, v > 0$$

(see, for example, Cox and Lewis, 1966, Chapter 2). The corresponding y -coordinates are, again independently, uniformly distributed on $(0, b)$. Note that this method automatically generates the events (x_i, y_i) in order of increasing x -coordinates, and terminates when the latest x -coordinate is greater than a .

4.5 Independence and random labelling

To assess the spatial association between the two types of events in a bivariate process, we can consider at least two different benchmark hypotheses :

- (i) *independence* - the two types of event are generated by a pair of independent univariate processes;
- (ii) *random labelling* - the two types of event are generated by labelling the events of a univariate process in a series of mutually independent Bernoulli trials.

These two hypotheses generate distinctively different K_{12} -functions.

Firstly, for any two independent processes of type 1 and type 2 events

$$K_{12}(t) = \pi t^2.$$

This follows from the fact that, if the two component processes are independent, then an event of type 1 has the same status, with respect to events of type 2, as an arbitrary point, hence the expected number of type 2 events within a disc of radius t centred on an arbitrary type 1 event is $\lambda_2 \pi t^2$, the expected number of type 2 events per unit area multiplied by the area of the disc. It follows that $K_{12}(t) = \pi t^2$ as claimed.

Secondly, for any randomly labelled process of type 1 and type 2 events,

$$K_{11}(t) = K_{22}(t) = K_{12}(t) = K(t), \quad (4.13)$$

where $K(s)$ is the K -function for the unlabelled univariate process. To see this, let $K(t)$ be the K -function of the unlabelled, univariate process consisting of all events, irrespective of type. Then, under random labelling the univariate processes of type 1 and type 2 events are each random thinnings of

the unlabelled process, and we have already seen in Section 4.2.1 that the K -function is invariant under random thinning, hence $K_{11}(t) = K_{22}(t) = K(t)$. Essentially the same argument shows that $K_{12}(t) = K(t)$.

Note that independence and random labelling are equivalent if and only if the component processes of type 1 and type 2 events are both Poisson processes. This makes it important to decide which, if either, is the natural benchmark of “no association” in a particular application.

4.6 Estimation of second-order properties

4.6.1 Stationary processes

For the reason given in Section 4.2, we shall focus initially on estimating the K -function.

In Section 4.2 we defined the function $K(t)$ by

$$\lambda K(t) = E[N_0(t)],$$

the expected number of further events within distance t of an arbitrary event, where the intensity λ is the mean number of events per unit area. An obvious estimator for λ is the observed number of events per unit area, $\hat{\lambda} = n/|A|$.

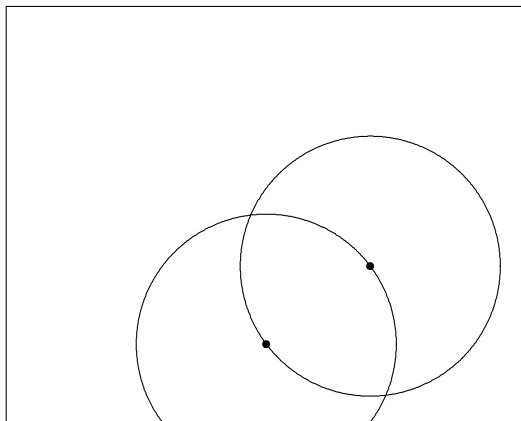
Similarly, because $E(t) = E[N_0(t)]$ is the expected number of further events within distance t of an arbitrary event, we can construct an estimator for $E(t)$ as follows. Let $r_{ij} = ||x_i - x_j||$. Define

$$\tilde{E}(t) = n^{-1} \sum_{i=1}^n \sum_{j \neq i} I(r_{ij} \leq t), \quad (4.14)$$

where $I(\cdot)$ denotes the indicator function.

The form of the estimator $\tilde{E}(t)$ in (4.14) suggests, correctly, that the K -function is closely connected to the distribution of inter-event distances, whose use in exploratory analysis we discussed in Section 2.2. However, $\tilde{E}(t)$ is negatively biased for $E(t)$ because of edge-effects. For a reference event within distance t of the boundary of A , the observed count of other events within distance t necessarily excludes any events which may have occurred within distance t but outside A . Several methods have been proposed to correct for this source of bias; see, for example, Stein(1991) or Baddeley (1999). The following method, which we shall use in all of our examples, is due to Ripley (1976).

Let $w(x, r)$ be the proportion of the circumference of the circle with centre x and radius r which lies within A . Write w_{ij} for $w(x_i, ||x_i - x_j||)$. Then, for any stationary isotropic process, w_{ij} is the conditional probability that an event is observed, given only that it is a distance $r_{ij} = ||x_i - x_j||$ away from

**FIGURE 4.1**

Construction of the edge-correction weights in Ripley's (1976) estimator for $K(t)$

the i th event x_i . See Figure 4.1, and note that in general $w_{ij} \neq w_{ji}$. Thus, an unbiased estimator for $E(t)$ is

$$\hat{E}(t) = n^{-1} \sum_{i=1}^n \sum_{j \neq i} w_{ij}^{-1} I(r_{ij} \leq t).$$

Finally, replacing the unknown intensity λ by $(n-1)/|A|$, we obtain Ripley's (1976) estimator for $K(t)$,

$$\hat{K}(t) = \{n(n-1)\}^{-1} |A| \sum_{i=1}^n \sum_{j \neq i} w_{ij}^{-1} I(r_{ij} \leq t). \quad (4.15)$$

In fact, Ripley used n^{-2} rather than $\{n(n-1)\}^{-1}$ in the expression for $\hat{K}(t)$; we prefer the given form for technical reasons although the distinction is clearly unimportant when n is large.

Ripley's estimator is approximately unbiased for sufficiently small t , the restriction on t being necessary because the weights w_{ij} can become unbounded as t increases. In practice this is not a serious problem. For example, when A is the unit square the theoretical upper limit of t is $\frac{1}{2}\sqrt{2} \approx 0.7$ but $\hat{K}(t)$ will seldom be required for such large values of t , partly because the sampling fluctuations in $\hat{K}(t)$ increase with t but also because it is not realistic to attempt to model effects which operate on the same physical scale as the dimensions of A .

The `splancs` package incorporates an algorithm written by Barry Rowl-

ingson for computing $w(x, y)$ when A is an arbitrary polygon. Explicit formulae for $w(x, u)$ can also be written down for simple shapes of region A , for example rectangular or circular, and these may be useful if computational efficiency is paramount. Suppose firstly that A is the rectangle $(0, a) \times (0, b)$. Write $x = (x_1, x_2)$ and let $d_1 = \min(x_1, a - x_1)$ $d_2 = \min(x_2, b - x_2)$; thus d_1 and d_2 are the distances from the point x to the nearest vertical and horizontal edges of A . To calculate $w(x, u)$ we need to distinguish two cases:

1. if $u^2 \leq d_1^2 + d_2^2$, then

$$w(x, u) = 1 - \pi^{-1} [\cos^{-1} \{ \min(d_1, u)/u \} + \cos^{-1} \{ \min(d_2, u)/u \}]; \quad (4.16)$$

2. if $u^2 > d_1^2 + d_2^2$, then

$$w(x, u) = 0.75 - (2\pi)^{-1} \{ \cos^{-1}(d_1/u) + \cos^{-1}(d_2/u) \}. \quad (4.17)$$

Note that (4.16) correctly gives $w(x, u) = 1$ when $u \leq \min(d_1, d_2)$. The above formulae apply to values of u in the range $0 \leq u \leq 0.5\min(a, b)$ which, as noted above, should be sufficient for practical purposes.

Now suppose that A is the disc with centre the origin and radius a . Let $r = \sqrt{(x_1^2 + x_2^2)}$ be the distance from x to the centre of the disc. Then, again distinguishing two cases, we have the following:

1. if $u \leq a - r$, then

$$w(x, u) = 1;$$

2. if $u > a - r$, then

$$w(x, u) = 1 - \pi^{-1} \cos^{-1} \{ (a^2 - r^2 - u^2)/(2ru) \}.$$

These formulae apply to values of u between 0 and a .

The sampling distribution of $\hat{K}(t)$ is analytically intractable, except in the case of a homogeneous Poisson process. Given any specific model and region A , the sampling distribution can be estimated by direct simulation. However, the theoretical expression for the variance of $\hat{K}(t)$ in a homogeneous Poisson process provides a useful benchmark in initial inspection of a plot of $\hat{K}(t)$. In what follows, we treat n , the number of events in A , as fixed.

For a homogeneous Poisson process, Ripley (1988) gives an asymptotic approximation to the sampling variance of $\hat{K}(t)$. Lotwick and Silverman (1982) give exact formulae whose evaluation in general requires extensive numerical integration, although they give explicit formulae for rectangular A . Chetwynd and Diggle (1998) give a different approximation, based on a thinning argument, that is easily computed for arbitrarily shaped A .

Ripley's asymptotic approximation, modified to take account of our non-standard choice of denominator in (4.15), is

$$v_R(t) = 2\{|A|/(n-1)\}^2 \{ \pi t^2/|A| + 0.96Pt^3/|A|^2 + 0.13(n/|A|)Pt^5/|A|^2 \} \quad (4.18)$$

where P denotes the perimeter of A . The approximation is accurate for relatively small values of t (see below).

Lotwick and Silverman's result, similarly modified and assuming rectangular A , is that the variance of $\hat{K}(t)$ is

$$v_{LS}(t) = \{n(n-1)\}^{-1} \{2b(t) - a_1(t) + (n-2)a_2(t)\} \quad (4.19)$$

where, for rectangular A with perimeter length P ,

$$b(t) = \pi t^2 |A|^{-1} (1 - \pi t^2 / |A|) + |A|^{-2} (1.0716 P t^2 + 2.2375 t^4),$$

$$a_1(t) = |A|^{-2} (0.21 P t^3 + 1.3 t^4)$$

and

$$a_2(t) = |A|^{-3} (0.24 P t^5 + 2.62 t^6).$$

These expressions are valid for t less than or equal to the shorter side-length of A .

Chetwynd and Diggle's approximation involves summations of functions of the edge-correction weights w_{ij} as follows. For any fixed t , define $\phi_{ij} = 0.5(w_{ij} + w_{ji})I(\|x_i - x_j\| \leq t)$. Further define

$$W_n = \sum_{i=1}^n \sum_{j \neq i} \phi_{ij},$$

$$X_n = \sum_{i=1}^n \sum_{j \neq i} \phi_{ij}^2$$

and

$$Z_n = \sum_{i=1}^n \left(\sum_{j \neq i} \phi_{ij} \right)^2.$$

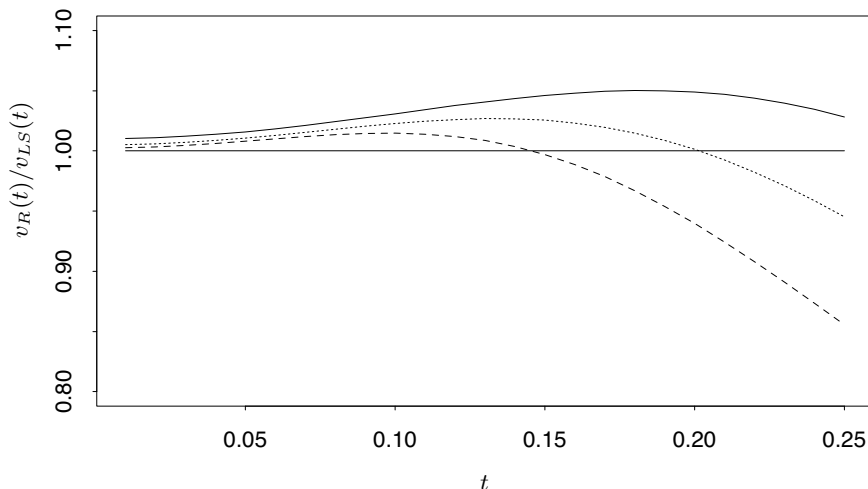
Write $n^{(k)} = n(n-1)\dots(n-k+1)$ and define $m_2(t) = X_n/n^{(2)}$, $m_3(t) = (Z_n - X_n)/n^{(3)}$ and $m_4(t) = (W_n^2 - 4Z_n + 2X_n)/n^{(4)}$. Then the estimated variance of $\hat{K}(t)$ is

$$v_{CD}(t) = (2|A|^2/n^{(2)}) \{ (3-2n)m_4(t) + 2(n-2)m_3(t) + m_2(t) \}. \quad (4.20)$$

Chetwynd and Diggle in fact give an explicit formula for $\text{Cov}\{\hat{K}(t), \hat{K}(s)\}$, of which (4.20) is a special case.

It is also useful to be able to assess the precision of an estimate of $K(t)$ without assuming a specific model. A simple way to do this is to subdivide A into equal sub-areas, estimate $K(t)$ separately within each sub-area and use the empirical variance over the separate estimates. Thus, if for each t we let k_i denote the estimate of $K(t)$ from the i th of m sub-areas, then our overall estimate is

$$\tilde{K}(t) = m^{-1} \sum_{i=1}^m k_i \quad (4.21)$$

**FIGURE 4.2**

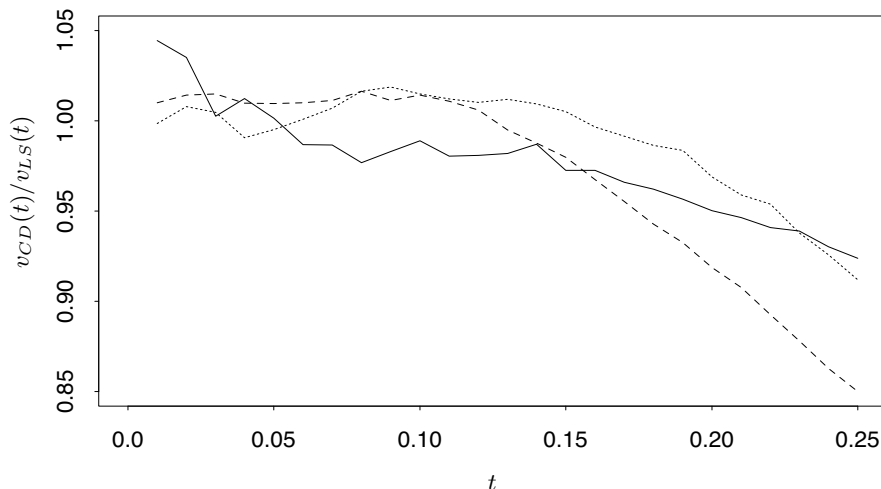
Comparison between Ripley's asymptotic approximation and the Lotwick-Silverman formula for the sampling variance of $\hat{K}(t)$. — : $n = 100$; - - - : $n = 200$; : $n = 400$. The horizontal line at height 1 corresponds to equality of the two formulae.

with approximate variance

$$\text{Var}\{\tilde{K}(t)\} \approx \{(m(m-1))\}^{-1} \sum_{i=1}^m \{k_i - \tilde{K}(t)\}^2. \quad (4.22)$$

The approximation in (4.22) arises for two reasons. Firstly, an element of approximation is inherent in using the sample variance of the k_i as an estimate of their true variance; secondly, dividing the sample variance of the k_i by m makes the implicit assumption that disjoint sub-regions give independent estimates k_i , which is correct for the homogeneous Poisson process, but not more generally. Furthermore, $\tilde{K}(t)$ can be expected to be less efficient than $\hat{K}(t)$ because it does not use information from pairs of events in different sub-regions; this may be an important consideration unless n , the number of events, is very large. All of these considerations suggest that the estimator $\tilde{K}(t)$ and its associated approximate variance should be used only for relatively small values of t , or when the artificial subdivision of A is replaced by genuine replication. We postpone further discussion of replicated patterns until Chapter 8.

We now illustrate the performance of the different estimators for $\text{Var}\{\hat{K}(s)\}$. We consider first the homogeneous Poisson process. Figure 4.2 shows the ratio of Ripley's asymptotic approximation (4.18) and the Lotwick-Silverman formula (4.19), for $n = 100, 200$ and 400 , and A the unit square. For small t , the approximation is excellent when $n = 100$ and, as would be expected, improves as n increases. At larger values of t , it is less reliable

**FIGURE 4.3**

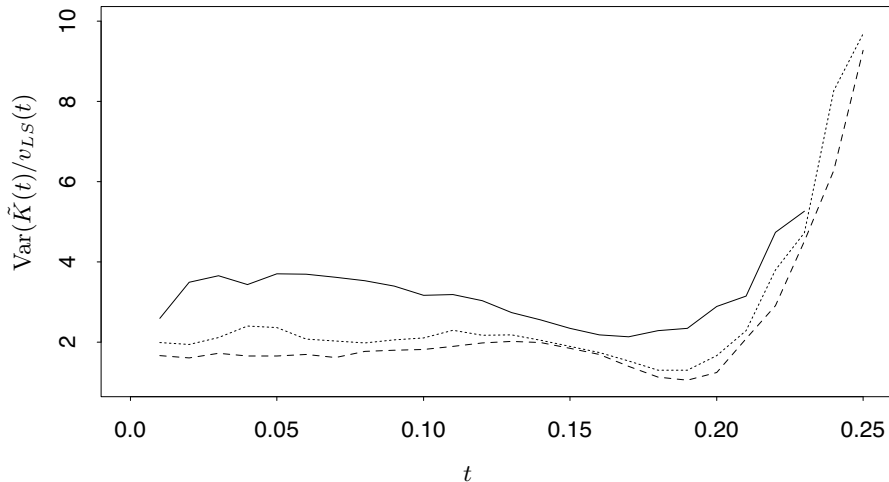
Comparison between Chetwynd and Diggle's estimator and the Lotwick-Silverman formula for the sampling variance of $\hat{K}(t)$. — : $n = 100$; - - : $n = 200$; : $n = 400$.

and does not necessarily improve as n increases. For a comparison between the Lotwick-Silverman formula and the Chetwynd-Diggle method, we need to simulate replicate patterns because the latter is a data-based estimate rather than a numerical approximation. Figure 4.3 is comparable to Figure 4.2 except that the comparison is now between the Chetwynd-Diggle estimated variance, averaged over replicate simulations, and the Lotwick-Silverman formula. The results point to a small negative bias in the Chetwynd-Diggle estimator at large distances.

In a second experiment, we again simulated homogeneous Poisson processes but now compared the two estimators $\hat{K}(t)$ and $\tilde{K}(t)$, the latter using a sub-division of the unit square into a 4×4 grid of smaller squares. The results summarised below are based on 1000 replicates of processes with each of $n = 100, 200$ and 400 events on the unit square, and for distances $t \leq 0.25$.

Recall that for the homogeneous Poisson process, the implicit assumption in (4.22) that disjoint sub-regions give independent estimates of $K(t)$ is correct, and the simulation results confirm this. Perhaps more interestingly, Figure 4.4 shows the ratio of the estimated variance of $\tilde{K}(t)$ and the Lotwick-Silverman formula $v_{LS}(t)$ for the variance of $\hat{K}(t)$. This confirms the inherent inefficiency of $\tilde{K}(t)$ relative to $\hat{K}(t)$, especially for small n and/or large t .

Finally, we simulated two non-Poisson processes, one generating aggregated patterns similar to Figure 1.2, the other regular patterns with a minimum permissible distance between events, similar to Figure 1.3. The results confirm that, as expected, formulae based on a Poisson assumption tend to

**FIGURE 4.4**

Comparison between the variances of $\tilde{K}(t)$ and of $\hat{K}(t)$. — : $n = 100$; : $n = 200$; --- : $n = 400$

under-estimate or over-estimate the variance of $\hat{K}(t)$ according to whether the underlying process is aggregated or regular, respectively. Specifically, in the aggregated case the estimated variances were between two and 12 times larger than the Poisson-based Lotwick-Silverman formulae, and in the regular case between three and seven times smaller over the range $t \leq 0.25$ (excluding small distances for which $\hat{K}(t)$ is identically zero because of the minimum permissible inter-event distance). However, the approximation (4.22) continued to give reasonable estimates for the variance of the less efficient estimator $\tilde{K}(t)$.

Our overall conclusions from these comparisons are the following. For the initial exploration of second-order structure of patterns which are close to completely random, it is useful to supplement a graphical display of $\hat{K}(t)$ with error bounds based on the Lotwick-Silverman formula for rectangular regions. For regions with an irregular boundary, such as arise typically in epidemiological applications, either the Chetwynd-Diggle formula can be used, or empirical bounds can be constructed from repeated simulations of a homogeneous Poisson process.

For patterns markedly different from completely random, Poisson-based approximations are unreliable. However, provided that a parametric model has been formulated, standard errors can again be estimated empirically from repeated simulations of the declared model. It is usually instructive, and straightforward in most modern computing environments, to supplement a graphical display of $\hat{K}(t)$ with envelopes of repeated simulations of a candi-

date model, both to quantify uncertainty and for an informal assessment of goodness-of-fit. We give examples in later chapters.

In the absence of a parametric model, the estimator $\tilde{K}(t)$ can be used if a reliable indication of precision takes precedence over efficiency of estimation.

4.6.2 Estimating the pair correlation function

Given an estimate $\hat{K}(t)$, a simple way to estimate the pair correlation function is to exploit the relationship between $K(t)$ and $\rho(t)$ by substituting the derivative of $\hat{K}(t)$ for $K'(t)$ on the right hand side of (4.4). Strictly, $\hat{K}(t)$ as defined at (4.15) is a step-function, and therefore non-differentiable. However, in practice we usually evaluate $\hat{K}(t)$ at a discrete set of equally spaced values of t , say $t = jh : j = 0, 1, 2, \dots, m$ for a suitably small value $h > 0$ and interpolate linearly. In this case, $\hat{K}'(t)$ is piece-wise constant,

$$\hat{K}'(t) = \{\hat{K}(jh) - \hat{K}((j-1)h)\}/h :: t \in ((j-1)h, jh),$$

leading to a histogram-like estimator for $\rho(t)$,

$$\hat{\rho}(t) = (2\pi t)^{-1} \hat{K}'(t). \quad (4.23)$$

Stoyan and Stoyan (1994) discuss a kernel-smoothed version, subsequently used by a number of authors including Møller, Syversveen and Waagepetersen (1998). For this, we again choose a positive value h , but also specify a univariate probability density function, $g(u)$ say, symmetric about $u = 0$, to define the smoothing weights. Stoyan and Stoyan's estimator for $\rho(t)$ is then

$$\tilde{\rho}(t) = |A|(2\pi t n^2)^{-1} \sum_{j \neq i} w_{ij}^{-1} h^{-1} g\{(t - u_{ij})/h\} \quad (4.24)$$

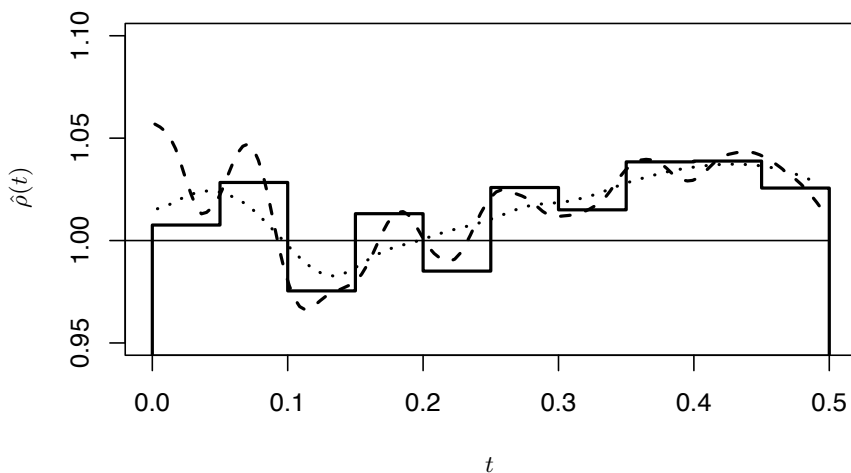
where, as earlier, $|A|$ is the area of A , $u_{ij} = ||x_i - x_j||$ and w_{ij} is Ripley's edge-correction. In the non-parametric smoothing literature, h is called the *band-width* and $g(u)$ the *kernel function*; see, for example, Silverman (1986).

Typically, h in (4.24) would usually be chosen somewhat larger than its counterpart when estimating $K(t)$. Another consideration is that (4.24) gives poor estimates of $\rho(t)$ at values of t close to zero; indeed, it gives positive estimates for negative t . A simple solution to this is to reflect the negative portion of the estimate in the y -axis, i.e. for each positive t , use the estimate

$$\hat{\rho}(t) = \tilde{\rho}(t) + \tilde{\rho}(-t), \quad (4.25)$$

although this can still leave substantial bias near $t = 0$ for aggregated patterns. Note also that the reflection in the y -axis does not arise for the histogram-like estimator (4.23), although the tendency to under-estimate $\rho(t)$ near $t = 0$ for aggregated patterns remains.

To illustrate these issues, we generated synthetic data by simulating a Poisson process with intensity $n = 500$ events on a unit square region A .

**FIGURE 4.5**

Comparison between histogram-like (solid line) and kernel smoothed estimates of the pair correlation function. Dashed and dotted lines correspond to kernel estimates using a quartic kernel and band-widths $h = 0.038$ and 0.076 , respectively.

Figure 4.5 shows estimates (4.23) and two versions of (4.25), based on the simulated data, and using a quartic kernel function,

$$g(u) = (15/16)(1 - u^2)^2 : -1 < u < 1,$$

in the smoothed estimator defined by (4.24) and (4.25). The histogram-like estimate uses bins of width 0.05, whilst the two versions of the kernel estimate use band-widths $h = 0.038$ and 0.076 . The smaller of the two band-widths induces an amount of smoothing comparable to the histogram-like estimate except near $u = 0$, apart from which the difference between these two is largely cosmetic, whereas the larger band-width gives a materially smoother estimate. Stoyan and Stoyan (1994) offer guidelines for choosing the band-width h in (4.24). The author's opinion is that attempting to optimise the choice of band-width is an over-elaboration of what is best seen as a simple exploratory device.

4.6.3 Intensity-reweighted stationary processes

If we assume that the process is intensity-reweighted stationary and, unrealistically, that the first-order intensity $\lambda(x)$ is known, we can estimate $K_I(t)$ by an easy modification of (4.18). The modification consists of re-scaling the inter-event distances by the product of the first-order intensities at the corre-

sponding two locations, leading to the estimator

$$\hat{K}_I(t; \lambda) = |A|^{-1} \sum_{i=1}^n \sum_{j \neq i} w_{ij}^{-1} I_t(u_{ij}) / \{\lambda(x_i) \lambda(x_j)\}. \quad (4.26)$$

This was proposed by Baddeley, Møller and Waagepetersen (2000), who went on to discuss the consequences of using an estimated first-order intensity $\hat{\lambda}(x)$ in place of the true $\lambda(x)$. Unsurprisingly, this turns out to be problematic, because of the difficulty of distinguishing empirically between non-constancy of $\lambda(x)$ and dependence between the events of the process. This relates to the equivalence of certain classes of Cox process and Poisson cluster process, to be discussed in Chapter 6. As a consequence, it is difficult in practice simultaneously to estimate non-parametrically both first-order and second-order properties of an intensity-reweighted stationary process. One situation in which it is easier to disentangle first-order and second-order properties is in design-based inference when independent replicate patterns can be assumed to have the same first-order intensity function. Another is in the analysis of case-control data in epidemiology. We discuss these topics in Chapters 5 and 9, respectively.

4.6.4 Multivariate processes

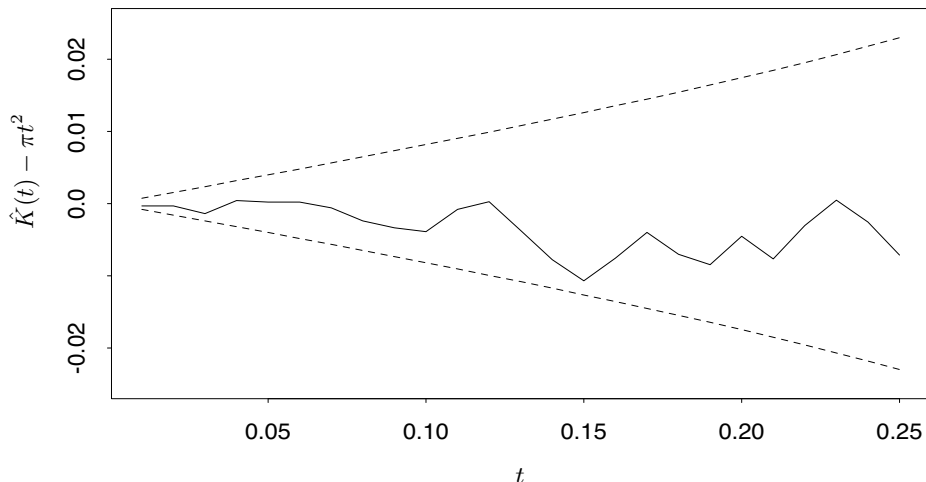
To estimate $K_{12}(s)$ for a bivariate pattern we use the same basic idea as in estimating $K(s)$, but measure distances between pairs of events of different types. Thus, if u_{ij} is the distance between the i^{th} event of type 1 and the j^{th} event of type 2, w_{ij} is as before, and the numbers of type 1 and type 2 events are n_1 and n_2 respectively, we can construct two estimates of $\hat{K}_{12}(s)$ as follows:

$$(i) \quad \hat{\lambda}_2 \tilde{K}_{12}(s) = n_1^{-1} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} w_{ij} I(u_{ij} \leq s)$$

$$(ii) \quad \hat{\lambda}_1 \tilde{K}_{21}(s) = n_2^{-1} \sum_{j=1}^{n_2} \sum_{i=1}^{n_1} w_{ji} I(u_{ij} \leq s)$$

We then combine the two estimates as a weighted average, to give

$$\begin{aligned} \hat{K}_{12}(s) &= (n_1 n_2)^{-1} |A| \left\{ n_1 \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} w_{ij} I(u_{ij} \leq s) \right. \\ &\quad \left. + n_2 \sum_{j=1}^{n_2} \sum_{i=1}^{n_1} w_{ji} I(u_{ij} \leq s) \right\} / (n_1 + n_2) \\ &= (n_1 n_2)^{-1} |A| \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} w_{ij}^* I(u_{ij} \leq s), \end{aligned} \quad (4.27)$$

**FIGURE 4.6**

The estimate $\hat{K}(t) - \pi t^2$ for the Japanese black pine data. — : data; — — : plus and minus two standard errors under complete spatial randomness.

where

$$w_{ij}^* = (n_1 w_{ij} + n_2 w_{ji}) / (n_1 + n_2).$$

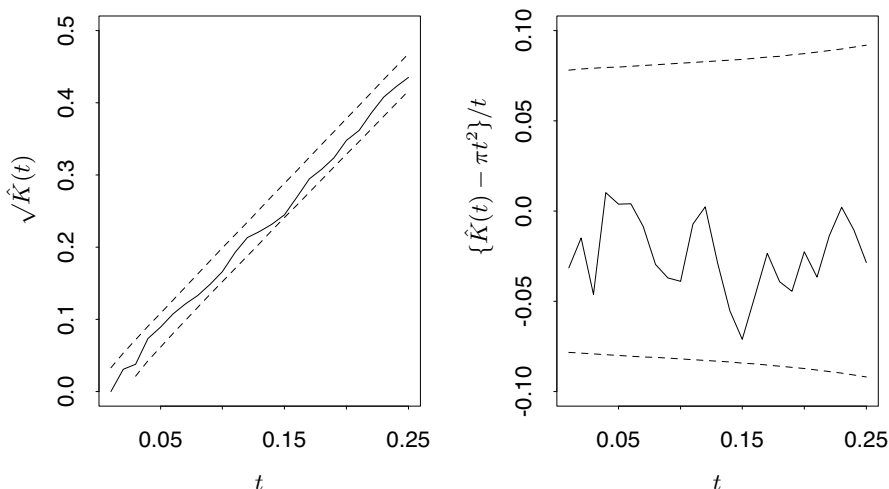
The variance formulae in Lotwick and Silverman (1982) include the bi-variate case. In addition to the functions $v(t)$, $a_1(t)$ and $a_2(t)$ defined above, let $c = n_2 / (n_1 + n_2)$. Then, when the component processes are independent homogeneous Poisson processes,

$$\begin{aligned} \text{Var}\{\hat{K}_{12}(t)\} &= (n_1 n_2)^{-1} |A|^2 [v(t) - 2c(1-c)a_1(t) + \\ &\quad \{(n_1 - 1)c^2 + (n_2 - 1)(1-c)^2\} a_2(t)] \end{aligned} \quad (4.28)$$

4.6.5 Examples

Figure 4.6 shows the estimate $\hat{D}(t) = \hat{K}(t) - \pi t^2$ for the Japanese black pine sapling data of Figure 1.1, together with plus and minus two standard deviation limits calculated from the Lotwick-Silverman formula under the assumption that the data are generated by a homogeneous Poisson process. Note that $\hat{D}(t)$ lies within these limits throughout the plotted range, suggesting compatibility with the Poisson assumption as in earlier analyses of these data.

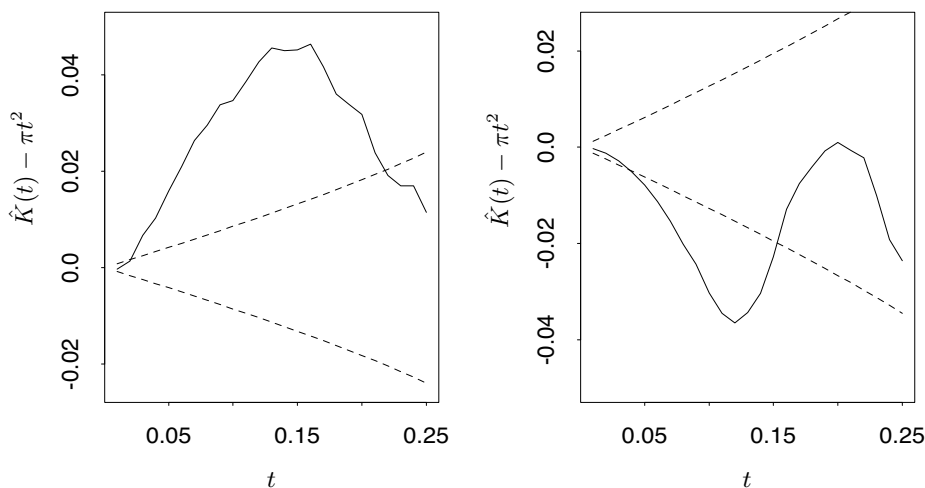
Note also that the standard deviation under the Poisson assumption is roughly linear in t . This is an indirect consequence of the Poisson quadrat count distribution, since $\hat{K}(t)$ is essentially an average of counts in circles of radius t , hence its mean and variance are both approximately proportional

**FIGURE 4.7**

Transformed estimates of $\hat{K}(t)$ for the Japanese black pine data. — : data; --- : plus and minus two standard errors under complete spatial randomness. The left-hand panel shows $\sqrt{\hat{K}(t)}$, the right-hand panel $\{\hat{K}(t) - \pi t^2\}/t$.

to t^2 . For this reason, some authors recommend plotting $\sqrt{\hat{K}(t)}$ against t to stabilise the sampling variance, and incidentally to linearise the plot under the Poisson assumption. We prefer to plot $\hat{D}(t)$ because of its direct physical interpretation in terms of counting numbers of events in circular regions; also, as we shall show in later chapters, plots of $D(t)$ can be used for preliminary estimation of parameters for some widely used models. If a variance-stable plot is required, either the square-root scale can be used, or a standardised difference, $\hat{D}(t)/t$, although the latter would become numerically unstable if extrapolated to $t = 0$. The two panels of Figure 4.7 show these two plots for the Japanese black pine data. In each case, the error limits are obtained by the appropriate transformation of the plus and minus two standard error limits for $\hat{K}(t)$ calculated by the Lotwick-Silverman formula.

The two panels of Figure 4.8 show $\hat{D}(t)$ with plus and minus two standard error limits under complete spatial randomness, for the redwood data of Figure 1.2, and for the cell data of Figure 1.3. In contrast to Figure 4.6, it is clear in both cases that the data are incompatible with complete spatial randomness, but for opposite reasons. As is now very familiar, the redwood data display strong spatial aggregation, and the cell data strong spatial regularity. The damped oscillatory behaviour of $\hat{D}(t)$ for the cell data is typical of regular patterns.

**FIGURE 4.8**

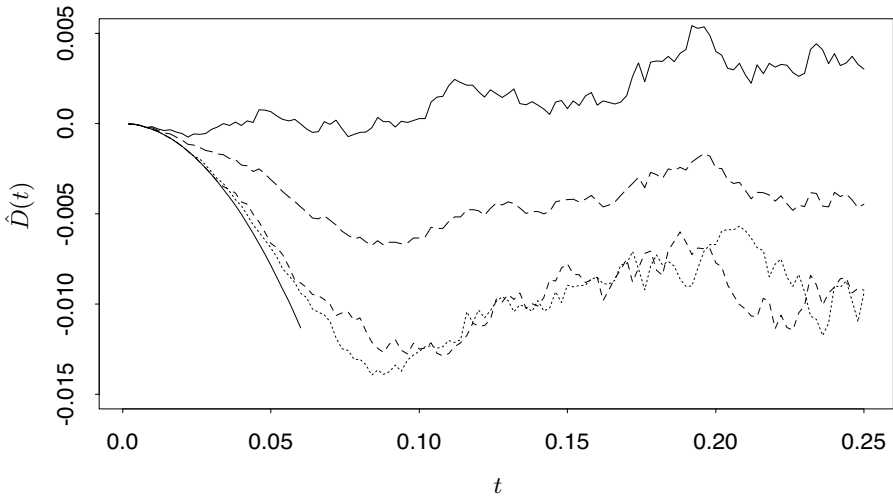
The estimate $\hat{D}(t) = \hat{K}(t) - \pi t^2$ for the redwood data (left-hand panel) and for the cell data (right-hand panel). — : data; - - - : plus and minus two standard errors under complete spatial randomness.

4.7 Displaced amacrine cells in the retina of a rabbit

The displaced amacrine cell data, shown in Figure 1.4, illustrate how estimated K -functions can be used in a specific context without explicit parametric modelling. The analysis presented here is adapted from Diggle (1985a).

The primary scientific interest in these data is to distinguish between two developmental hypotheses (Hughes, 1981). Recall that the two types of cell are those which respond to a light being switched *on* or *off*, respectively. The *separate layer* hypothesis is that the on and off cells are initially formed in two separate layers which later fuse to form the mature retina, whilst the *single layer* hypothesis is that the two types of cell are initially undifferentiated in a single layer and acquire their separate functions at a later stage.

Figure 4.9 shows estimates of $K_{ij}(t)$ for each of $(i, j) = (1, 1)$, $(1, 2)$ and $(2, 2)$, where types 1 and 2 refer to on and off cells, respectively, and an estimate of $K(t)$ for the superposition of both types of cell. Note firstly that the estimate of $K_{12}(t) - \pi t^2$ is close to zero throughout the plotted range. We could use the Lotwick-Silverman formula to compute the sampling variance of $K_{12}(t)$ under the assumption that the two types of cell form independent Poisson processes, but this would likely over-estimate the variance because, as shown by the estimates of $K_{11}(t)$ and $K_{22}(t)$, the components' patterns are markedly more regular than the Poisson process. However, we can test the

**FIGURE 4.9**

Second-order properties of displaced amacrine cells. Functions plotted are $\hat{D}(t) = \hat{K}(t) - \pi t^2$ as follows: - - - : on cells; : off cells; — — — : all cells; : bivariate. The parabola $-\pi t^2$ is also shown as a solid line.

independence hypothesis using a procedure also suggested by Lotwick and Silverman (1982). If we wrap the rectangular observation window A onto a torus, independence of the component processes would imply that the sampling distribution of $\hat{K}_{12}(t)$ is invariant to a random toroidal shift of either pattern. This in turn implies that we can conduct a Monte Carlo test of independence by comparing the value of a suitable test statistic for the observed data with values generated under a sequence of independent random toroidal shifts. Using the test statistic

$$u = \sum_{k=1}^{125} t_k^{-2} \{ \hat{K}_{12}(t_k) - \pi t_k^2 \}^2,$$

where $t_k = 0.002, 0.004, \dots, 0.250$, we implemented a Monte Carlo test by recalculating u after each of 99 independent random toroidal shifts, and obtained a p -value of 0.12. Thus, at least with regard to their second-order properties, the evidence against the hypothesis of independent components is weak. This is supportive of the separate layer hypothesis. However, the evidence is not yet conclusive since the result that $K_{12}(t) = \pi t^2$ is necessary, but not sufficient, for independence.

We now examine $\hat{K}_{11}(t)$ and $\hat{K}_{22}(t)$, and note that they are very similar to each other, but markedly different from $\hat{K}(t)$ for the superposition. This suggests that if the observed pattern is the result of a labelling of an initially

undifferentiated set of cells, then the labelling cannot be random. We could formally test the hypothesis of random labelling, but this seems unnecessary in view of the very clear difference between $\hat{K}(t)$ for the superposition and the two very similar estimates $\hat{K}_{11}(t)$ and $\hat{K}_{22}(t)$.

The argument so far leaves open the possibility that the observed pattern is the result of a non-random labelling process. However, note that both $\hat{K}_{11}(t)$ and $\hat{K}_{22}(t)$ are zero for $t < 0.025$ approximately, whereas $\hat{K}(t)$ for the superposition is non-zero at much smaller values of t . For this to arise from a labelling process, the following would have to be true: in the undifferentiated process, close pairs of cells are allowed but mutually close triples are forbidden; and in the labelling process, the two members of a close pair must always be oppositely labelled. This seems implausible, and the analysis therefore points strongly towards the separate layer hypothesis being the correct explanation. Subsequent work by Eglin, Diggle and Troy (2005) and Diggle, Eglen and Troy (2006) led to a refinement of the separate layer hypothesis; in Section 8.3.6 we shall re-visit this example accordingly.

4.8 Estimation of nearest neighbour distributions

We now consider estimation of the two nearest neighbour distribution functions introduced in Sections 2.3 and 2.4. These are: $F(x)$, the probability that the distance from an arbitrary point to the nearest event is less than or equal to x ; and $G(x)$, the probability that the distance from an arbitrary event to the nearest other event is less than or equal to x .

In either case, the simplest estimator is the empirical distribution function of observed nearest neighbour distances, as used in Sections 2.3 and 2.4. These estimators are biased because of edge-effects. The bias does not affect the validity of a Monte Carlo test of complete spatial randomness, or indeed of any other specified model, but would be problematic if we were directly concerned with estimation.

One approach to edge-correction is the following, due to Ripley (1977). Consider first an estimator for $G(y)$. Let $(y_i, d_i) : i = 1, \dots, n$, denote the distances from each event to the nearest other event in A and to the nearest point on the boundary of A , respectively, and define the estimator $\tilde{G}(y)$ to be the proportion of nearest neighbour distances $y_i \leq y$ amongst those events at least a distance y from the boundary of A ; thus

$$\tilde{G}(y) = \#(y_i \leq y, d_i > y) / \#(d_i > y).$$

To estimate $F(x)$, similarly let $(x_i, e_i), i = 1, \dots, m$, denote the distances from each of m sample points to the nearest event in A and to the nearest point on the boundary of A , and define

$$\tilde{F}(x) = \#(x_i \leq x, e_i > x) / \#(e_i > x).$$

As in Section 2.4, for estimation of $F(x)$ we would suggest locating the m sample points in a regular grid. Alternatively, we could avoid altogether the need for a grid of sample points by computing directly the areal proportion of the study region for which the distance to the nearest event is less than or equal to x , as described by Lotwick (1981).

A technically simpler procedure for dealing with edge-effects would be to use the buffer-zone method, taking measurements only from points or events sufficiently far from the boundary of A , so that edge effects do not arise for the range of distances of interest. As noted in Section 1.3, the obvious disadvantage of this procedure is that it effectively throws away a non-trivial proportion of the data. Recall also from Section (2.7) that Baddeley and van Lieshout's J -function, defined by $J(x) = \{1 - G(x)\} / \{1 - F(x)\}$, can be estimated reliably without any correction for edge-effects.

The sampling distributions of estimators for $F(\cdot)$ and $G(\cdot)$ appear to be intractable, although for large data-sets the device of splitting the observation region A into sub-regions to form pseudo-replicates is available, whilst simulation can be used to assess the fit of a parametric model.

4.8.1 Examples

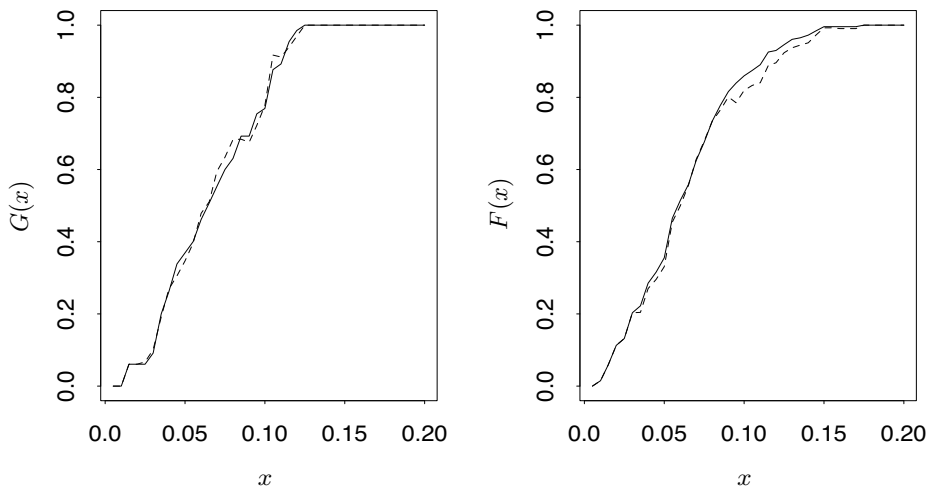
Figure 4.10 compares the empirical distribution functions, $\hat{G}(\cdot)$ and $\hat{F}(\cdot)$, with the edge-corrected estimators $\tilde{G}(\cdot)$ and $\tilde{F}(\cdot)$ for the Japanese black pine sapling data of Figure 1.1. Notice that the two estimates in each case are very similar, but that the edge-corrected estimators are not necessarily monotone.

Figures 4.11 and 4.12 are the corresponding plots for the redwood seedling data of Figure 1.2 and for the cell data of Figure 1.3. For comparability, all six plots in Figures 4.10, 4.11 and 4.12 evaluate the estimates over the same set of distances. The qualitative differences amongst these three data-sets are again clear.

4.9 Concluding remarks

The K -function, and the two nearest neighbour distribution functions $F(\cdot)$ and $G(\cdot)$, provide complementary tools for the description of spatial point processes.

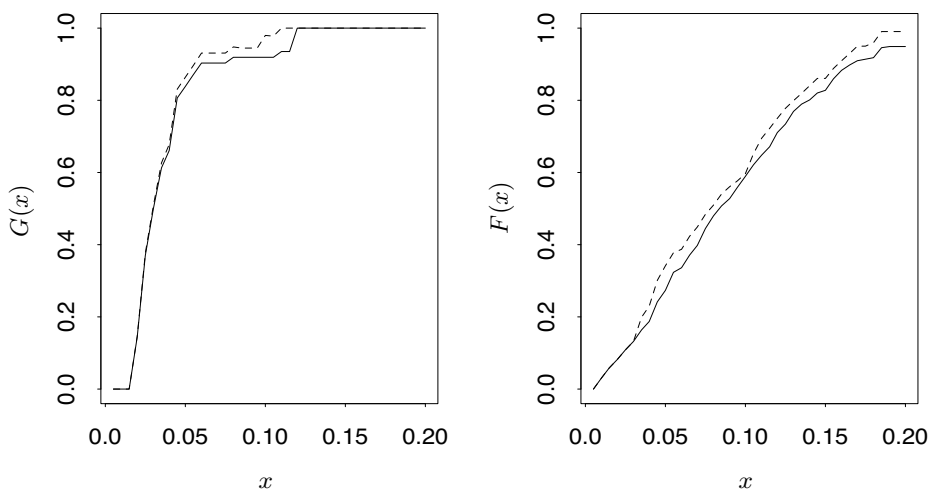
The K -function is the most amenable of the three to theoretical analysis. We shall see in later chapters that its algebraic form can be derived for a number of useful models. Its physical interpretation as a scaled expectation is also a useful property. The sampling distribution of its estimator $\hat{K}(t)$ is reasonably well understood in the case of a homogeneous Poisson process,

**FIGURE 4.10**

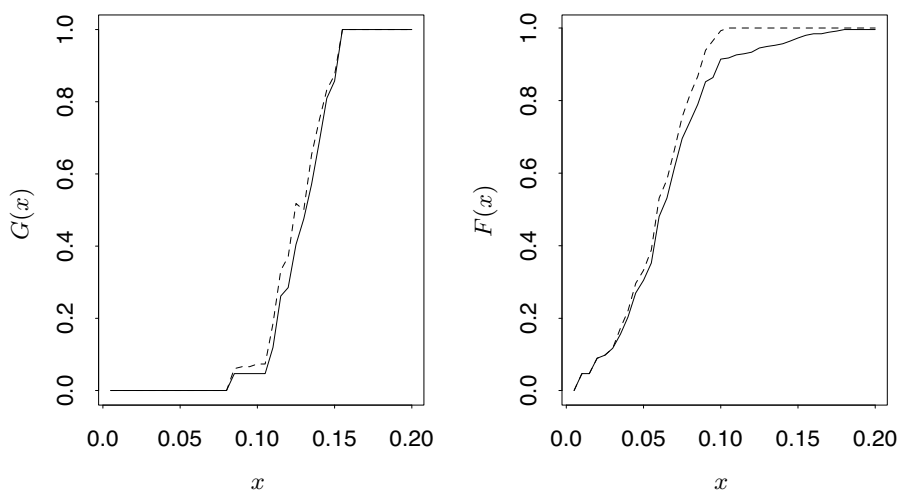
Estimates of $F(x)$ (left-hand panel) and of $G(x)$ (right-hand panel) for the Japanese black pine data. — : $\hat{G}(\cdot), \hat{F}(\cdot)$; - - : $\tilde{G}(\cdot), \tilde{F}(\cdot)$

but otherwise must be assessed by splitting the observation region into sub-regions, or by simulation. The choice between using estimates of $K(t)$ or $\rho(t)$ to summarise the second-order properties of an observed pattern is to some extent a matter of taste. **One clear advantage in estimating $K(t)$ rather than $\rho(t)$ is that this avoids the need to choose a bin-width or band-width.** In the author's opinion, this is conclusive when the number of events is relatively small, say of the order of one hundred or less. For observed patterns with larger numbers of events, the choice is less clear-cut.

Estimates of all three functions, $K(\cdot)$ (or $\rho(\cdot)$), $F(\cdot)$ and $G(\cdot)$ are useful for exploratory assessment of departure from complete spatial randomness. More generally, we shall see in later chapters that the estimated K -function is a useful and versatile tool. In the author's opinion, the principal use of estimates of $F(\cdot)$ and $G(\cdot)$ is to provide goodness-of-fit measures that are complementary to the second-order description provided by the K -function. In this respect, it is worth noting that the function $J(x) = \{1 - G(x)\} / \{1 - F(x)\}$ can sometimes be calculated explicitly when its separate components $F(x)$ and $G(x)$ are intractable.

**FIGURE 4.11**

Estimates of $F(x)$ (left-hand panel) and of $G(x)$ (right-hand panel) for the redwood data. — : $\hat{G}(\cdot), \hat{F}(\cdot)$; - - - : $\tilde{G}(\cdot), \tilde{F}(\cdot)$

**FIGURE 4.12**

Estimates of $F(x)$ (left-hand panel) and of $G(x)$ (right-hand panel) for the cell data. — : $\hat{G}(\cdot), \hat{F}(\cdot)$; - - - : $\tilde{G}(\cdot), \tilde{F}(\cdot)$

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